Federated Full-Envelope LPV Identification from Incomplete Local Coverage

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***Abstract*—Individual vehicles may supply calibration data from only a narrow speed region, although the fleet collectively covers the operating interval. Pooling those records can recover missing scheduling information but also mix incompatible vehicle dynamics. We investigate compatible federated identification of linear-parameter-varying (LPV) maps under this joint coverage and heterogeneity limitation. With available lateral states, clients identify local matrices and exchange additive regression statistics and assignment scores to learn shared group maps. A nonlinear vehicle benchmark separates scheduling-rank deficiency from model mismatch and finite-data error. Across ten label-free fleets, grouped fitting reduces mean paired full-envelope matrix error by 93.41% relative to restricted Local and 23.74% relative to Global. At locally unobserved speeds, mean error is 3.527%, compared with 65.568% for Local and 4.388% for Global. The distributed implementation reproduces centralized grouped fitting. The richer locally regularized comparator retains essentially the same error as restricted Local, and all 18 selected groups have full-rank scheduling designs. Separate scheduled-LQI tests provide modest supporting tracking gains. The validated setting is controlled equilibrium-centered calibration with available states, exact current regressors, and noisy next-state targets.**

***Index Terms*—federated identification, LPV systems, incomplete operating coverage, heterogeneous vehicles**

I. Introduction

Speed-dependent lateral dynamics and persistent vehicle differences create two distinct identification problems. An LPV representation captures operating variation through a measured scheduling variable, while different vehicles may require different coefficient maps. A fleet-wide model can average incompatible dynamics, whereas independent identification requires each vehicle to excite enough operating conditions.

LPV identification through basis-dependent regression and scheduling excitation is established [1], [2]. Local approaches first identify models at fixed operating points and then construct a scheduling-dependent map [3]. Here the common physical state coordinates are available, so local models do not require alignment of independently identified latent realizations. The information limitation is instead incomplete scheduling coverage: each client observes only a contiguous speed block. Repeating transitions at those same speeds reduces estimation noise but cannot supply missing basis directions. Complementary fleet coverage can supply them, provided that the combined clients have sufficiently compatible dynamics. Figure 1 summarizes this coverage limitation and the client–server identification procedure.

Federated learning exchanges computed information while retaining records locally [4]. Clustered learning addresses persistent client heterogeneity by learning several models [5]. Our earlier vehicle-identification work studies uncertainty-aware grouping for controller design [6]. The present question is whether clients whose scheduling regressions are individually underdetermined can jointly identify group LPV maps across the speed interval. Its primary endpoint is recovery across incomplete operating coverage, with control as a downstream check. Additive regression and clustering are established components rather than independent novelty claims.

The contribution is a controlled study and distributed construction for this joint information limitation. First, blocked-speed basis selection distinguishes missing scheduling information from a physical representation that already extrapolates adequately. Second, clients participate in hard-assignment group regression through local statistics and scores, without identifying full local coefficient maps or supplying family labels. Third, paired fleet evaluation separates full-envelope, observed-speed, and unobserved-speed error and checks the resulting models in nonlinear scheduled-LQI tracking. Numerical centralized–federated agreement verifies the implementation, not a new statistical advantage of federation over the same centralized records.

II. Identification Problem

A. Available states and incomplete speed coverage

Client i has lateral states $x = [\beta, r]^\top$, steering input u, and longitudinal speed v. The benchmark uses radians for β, u and rad/s for r. At a fixed cornering equilibrium, local perturbations satisfy approximately

$$\Delta x_{i,k+1} = A_i(v)\Delta x_{i,k} + B_i(v)\Delta u_{i,k} + e_{i,k}, \qquad (1)$$

where A_i, B_i are discrete-time matrices and $e_{i,k}$ includes nonlinear approximation error and next-state observation noise. Define $m_i(v) = [A_{11}, A_{12}, A_{21}, A_{22}, B_1, B_2]$. A group g_i, unknown during fitting, approximates the client map as

$$m_i(v) \approx \phi(v)^\top \Theta_{g_i}, \qquad \Theta_g \in \mathbb{R}^{L\times 6}. \qquad (2)$$

Within-group variation remains. Each client provides matrix estimates at speeds in $\mathcal{V}_i \subset \mathcal{V} = [10, 30]$ m/s. "Full envelope" means this speed interval along the prescribed cornering curve, rather than an arbitrary nonlinear state/input domain.

B. Fixed representation and local estimation

The scheduling span is

$$b(v) = [1, v^{-1}, v^{-2}, z, z^2, z^3, z^4]^\top, \quad z = (v-20)/10. \quad (3)$$

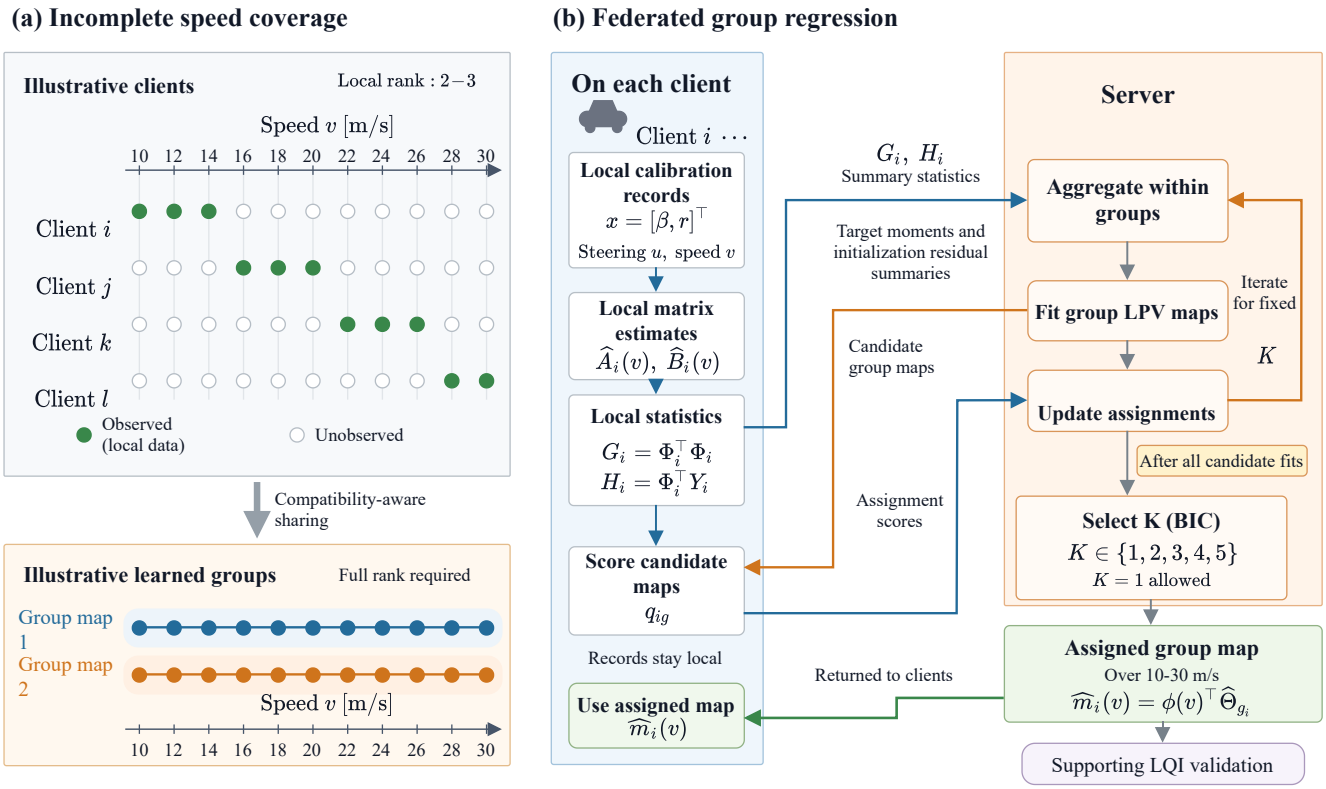


Fig. 1. Overview of compatible federated LPV identification. (a) Illustrative clients observe narrow speed blocks. Their scheduling designs have rank two or three for a seven-function basis; compatible sharing can supply missing directions if the pooled group design has rank seven. The displayed groups are illustrative. (b) Clients retain calibration records and local matrix targets, exchange additive statistics, summaries, and assignment scores, and receive fitted group maps. Fitting and assignment iterate for each candidate K, followed by BIC-based selection. Available states and equilibrium-centered calibration are assumed. The envelope is the 10–30 m/s interval along the prescribed cornering curve; LQI provides a supporting usability check.

Let the rows of B_{ref} be $b(v)^\top$ on a fixed 401-point uniform 10–30 m/s grid, with thin QR factorization $B_{\text{ref}} = QR$. The implemented row basis is $\phi(v)^\top = b(v)^\top R^{-1}$. This invertible transform changes conditioning, not the represented span. Development-only blocked-speed validation selects $L = 7$ for cornering, while the straight-line negative control retains $[1, v^{-1}, v^{-2}]$. No control comparison selects the representation.

At each observed speed, a client stacks 36 independent transitions. With rows $[\Delta x_k^\top, \Delta u_k]$ in Z and next-state rows in X_+, it estimates

$$\widehat{C} = \arg\min_C \|X_+ - ZC\|_F^2, \qquad [\widehat{A}, \widehat{B}] = \widehat{C}^\top. \tag{4}$$

Current state/input regressors are exact, and only next-state targets receive noise. These estimates, with their speeds, are the group-regression targets. Vehicle parameters, exact evaluation matrices, and family labels do not enter that fitting or assignment interface.

The generator computes each centering equilibrium from simulator parameters. Thus available states alone do not imply identification from ordinary driving records. A controlled experiment could instead measure an operating-point center, but that alternative is not validated here. The same clients contribute calibration records and receive the maps. Evaluation at additional speeds tests operating-condition generalization; fresh fleets repeat the whole procedure rather than testing a frozen library on wholly new recipients.

III. Federated Compatible Regression

A. Additive fitting and conditional identifiability

Client i forms $\Phi_i \in \mathbb{R}^{n_i \times L}$ and matrix-entry targets $Y_i \in \mathbb{R}^{n_i \times 6}$. It contributes $G_i = \Phi_i^\top \Phi_i$ and $H_i = \Phi_i^\top Y_i$. For a fixed partition, the group fit is

$$\widehat{\Theta}_g = \left(\sum_{i:g_i=g} G_i + (\lambda + \epsilon) I \right)^{-1} \sum_{i:g_i=g} H_i, \tag{5}$$

with $\lambda = 10^{-8}$ and numerical diagonal $\epsilon = 10^{-12}$. The same unweighted ridge objective and basis are used in centralized and federated grouped fitting. The matrix estimates have equal weight; local estimation covariance is not propagated. Unequal noise or excitation could therefore motivate weighting, but is outside the present comparison.

Proposition 1: For a fixed partition and exact shared-map targets $Y_i = \Phi_i \Theta_g$, the unregularized coefficients are unique if and only if the vertically stacked group design has rank L, equivalently $\operatorname{rank}(\sum_{i:g_i=g} G_i) = L$.

Justification: The aggregate Gram matrix is positive definite exactly when the stacked design has no nonzero null direction.

TABLE I
FEDERATED GROUP-REGRESSION PROCEDURE.

Inputs: Local speed/matrix pairs, fixed basis, candidate $K = 1, \dots, 5$, minimum group size five.
1. Aggregate local statistics and target moments. Fit Global, broadcast it, and collect residual summaries.
2. For each K, use one initialization for $K = 1$ or twelve seeded K-means initializations otherwise.
3. Alternate group-statistic solves, map broadcasts, local score returns, and hard assignments. Stop at unchanged assignments or 30 iterations. Discard initializations with groups smaller than five.
4. Refit the final admissible partition if needed. Retain the lowest-BIC initialization for each K, then the lowest-BIC K.
Outputs: Group maps and client assignments, including $K = 1$.

Otherwise, adding a null-direction coefficient matrix leaves all observed predictions unchanged. Additivity also makes the centralized and distributed solutions identical under the same numerical conventions.

This statement is conditional on a partition and model class. It neither establishes correct group discovery nor removes pooling bias. Ridge gives a numerical answer in deficient groups but does not restore information.

B. Assignment and model selection

The server derives a common entry scale from additive counts, sums, and squared sums of Y_i. For entry j, let s_j be its pooled standard deviation, floored at 0.05 times the median standard deviation, and set $D = \operatorname{diag}(s_1, \dots, s_6)$. Clients return scores

$$q_{ig} = \|(Y_i - \Phi_i \widehat{\Theta}_g) D^{-1}\|_F^2, \qquad g_i \leftarrow \arg\min_g q_{ig}. \tag{6}$$

The common scaling balances entries during assignment. Fitting remains entry-wise ridge regression. Initial groups come from K-means on standardized six-entry client mean residuals around the Global fit. The server receives those summaries, not the local matrices themselves.

The model-selection score uses the normalized assignment residuals:

$$\mathrm{BIC} = n \log\big(\max\{\mathrm{SSE}/n, 10^{-12}\}\big) + p_K \log n, \tag{7}$$

where $\mathrm{SSE} = \sum_i q_{i,g_i}$, $n = 6\sum_i n_i$, and $p_K = 6LK + K - 1$. Ties choose the first group index. This is a heuristic selection criterion for finite noisy matrix targets and within-group variation. It does not certify a likelihood model, global convergence, or physical-family recovery. Minimum group size does not imply full scheduling rank, so group fit and group information must be assessed separately.

C. Communication

For $L = 7$, each local Gram has 28 distinct entries and its right-hand side has 42. Target moments, initialization summaries, group tags, assignment scores, and model broadcasts also enter the accounting. Including all candidate orders and restarts, mean label-free traffic is 2.243 MB per fleet at eight bytes per scalar. This exhaustive search is not shown to be communication-efficient. The emulator retains raw records locally, but informative statistics and scores are shared. No secure aggregation or differential privacy is implemented.

TABLE II
VEHICLE CENTERS USED ONLY TO GENERATE THE FLEET. ALL CLIENTS HAVE $l_f = 1.2$ M AND $l_r = 1.6$ M.

Center	m [kg]	I_z [kg m^2]	C_f	C_r
Nominal	1500	2500	80000	80000
Heavy	1725	3000	84000	80000
Handling	1425	2250	72000	88000

Stiffnesses in N/rad.

IV. BENCHMARK AND EVALUATION

A. Nonlinear vehicles and calibration records

The plant is a two-state single-track model with nonlinear tanh tires:

$$\dot{\beta} = (F_f + F_r)/(mv) - r, \tag{8}$$

$$\dot{r} = (l_f F_f - l_r F_r)/I_z, \tag{9}$$

$$F_a = \mu F_{za} \tanh(C_a \alpha_a/(\mu F_{za})), \quad a \in \{f, r\}. \tag{10}$$

Here $\alpha_f = u - \beta - l_f r/v$, $\alpha_r = -\beta + l_r r/v$, $F_{zf} = mgl_r/(l_f + l_r)$, $F_{zr} = mgl_f/(l_f + l_r)$, $g = 9.81$ m/s^2, and $\mu = 0.9$. Each fleet has ten clients from each center in Table II. Mass and inertia receive independent uniform multiplicative scatter of $\pm 2.5\%$ and axle stiffnesses $\pm 5\%$. Geometry remains fixed. Family labels are withheld from primary block allocation and fitting.

Calibration follows curvature 0.005 m^{-1} and uses RK4 integration with sampling time 0.01 s. Zero-mean independent Gaussian perturbation standard deviations are approximately 0.25° for β, 2°/s for r, and 0.30° for steering. Added next-state noise has standard deviations 0.01° and 0.03°/s. These are independently initialized one-step records, not successive samples from a driving trajectory.

The four local speed blocks are $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, and $\{28, 30\}$ m/s. Globally balanced random allocation assigns them to eight, eight, seven, and seven clients, respectively, without consulting family labels. Primary fleet seeds are 371–380. Development representation selection uses separate seeds 301–310, contiguous blocked-speed validation, a one-standard-error rule, and a practical relative-error floor of 10^{-4}.

B. Endpoints, averaging, and baselines

Reference matrices $m_i^\star(v)$ are numerical continuous-time Jacobians at the prescribed equilibrium, discretized by an augmented matrix exponential. They are used for development representation selection and evaluation, not finite-data group fitting. The primary relative error is

$$e_i(v) = \frac{\|\widehat{m}_i(v) - m_i^\star(v)\|_2}{\|m_i^\star(v)\|_2}, \qquad E_s = \frac{1}{30 \cdot 11} \sum_i \sum_{v \in \mathcal{V}_{\mathrm{grid}}} e_i(v). \tag{11}$$

where $\mathcal{V}_{\mathrm{grid}} = \{10, 12, \dots, 30\}$ m/s. The metric uses the stated physical coordinates without entry normalization. It is a fixed-coordinate benchmark error, not a coordinate-invariant measure

or a tracking-performance surrogate. Observed/unobserved-speed summaries average only their respective client-speed pairs within each fleet. Reported errors average E_s over ten independent fleets. For method M and baseline B, reported gains average $100(1 - E_s^M/E_s^B)$ over paired fleets, rather than taking the ratio of grand mean errors. Confidence intervals resample those ten fleet units 20,000 times.

Restricted Local fits $L = 3$ with ridge 10^{-8} using each client's records. Global, centralized groups, and federated groups use $L = 7$ and the same fleet records. Thus Local tests access to complementary information, while Global tests sharing without compatibility selection. Centralized groups isolate the distributed implementation from the model benefit. Regularized Local uses $L = 7$ with a ridge penalty toward the client's $L = 3$ fit, expressed in the richer basis. The penalty strength is selected by leave-one-speed-out validation from $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 10^{2}\}$; clients with only two speeds use the fixed middle candidate 10^{-2}. This comparator requires no shared records. Its matched check and the group-information diagnostics reuse the original ten label-free fleets and calibration protocol. Reconstructed parent errors were verified against the saved client-speed results. These checks provide no additional independent confirmation fleets. An additional full-coverage recipient reference is more accurate than family pooling at approximately equal transition budget: sharing addresses missing local access, not inherent superiority at equal recipient information.

V. Identification Results

A. Why complementary coverage is needed

At curvature 0.005 m^{-1}, development blocked-speed matrix error falls from 2.6607% at order three to 0.0007% at order seven. Straight-line selection retains order three, avoiding unnecessary richness when the physical span is already adequate. The four local designs have ranks three, three, three, and two for seven coefficients. Their union has rank seven and audited condition 3.406. This shows the potential to add missing directions; it does not assert that every learned group inherits full union coverage.

With finite transitions and known compatibility, oracle-family pooling achieves 1.816% unobserved-speed error, versus 66.121% for Local and 4.296% for Global on separate development fleets. The richer locally regularized comparator does not improve on Local there. These oracle results motivate the mechanism before removing labels, but are not the proposed federated result.

B. Primary label-free results

Table III separates the full-envelope endpoint from its observed and unobserved regions. Mean paired gains over Local and Global are 93.41% and 23.74%, respectively. The Global-relative 95% interval is $[15.39, 31.04]\%$, with improvement in eight of ten fleets. Two groups are selected in eight fleets and one in two. Restricted Local is most accurate at its own observed speeds, whereas sharing substantially reduces its unsupported extrapolation error. Grouped sharing additionally reduces Global's pooling error. Figure 2 shows where those changes occur. Aggregate improvement does not extend to every client: 71 of 300 client instances have larger unobserved-speed error than Global, after excluding numerical ties.

TABLE III
Mean relative matrix error [%] across ten label-free fleets. Restricted Local uses $L = 3$; Regularized Local and shared methods use $L = 7$. Regularized Local is penalized toward its own $L = 3$ fit.

Method	Full	Observed	Unobserved
Restricted Local	49.562	1.930	65.568
Regularized Local	49.562	1.930	65.568
Global	4.250	3.840	4.388
Centralized groups	3.249	2.423	3.527
Federated groups	3.249	2.423	3.527

The richer locally regularized fit has negligible effect on Local's full-envelope error at the reported precision. At unobserved speeds, it reduces mean error by only 0.00016 percentage points. Grouped fitting improves full-envelope error over Regularized Local by 93.41%, with a 95% paired fleet-bootstrap interval of $[92.59, 94.21]\%$. The corresponding unobserved-speed gain is 94.59%, interval $[93.92, 95.25]\%$, with wins in all ten fleets for both endpoints. Thus the sharing benefit persists against this specified richer local alternative. This comparison does not establish failure of all local estimators or stronger physical priors.

Federated and centralized protocols select identical partitions and group counts in all 20 fleet/protocol jobs, including the separate family-stratified coverage reference. Maximum coefficient discrepancy is 3.20×10^{-14} and maximum BIC discrepancy is 1.87×10^{-9}. This is implementation verification under matched inputs.

C. Scheduling information in selected groups

All 18 selected groups contain observations at all eleven grid speeds and have scheduling-design rank seven. Their stacked design condition numbers range from 3.26 to 3.87 in the fixed QR basis. Every individual seven-function design remains deficient, with rank two or three. The learned partitions therefore satisfy the scheduling-information condition in Proposition 1 for this benchmark. These condition numbers refer to the scheduling design, not its Gram matrix or the complete dynamical-identification information matrix. Full rank does not remove within-group bias or guarantee physical-family recovery.

A known-family reference loses a speed block in one fleet and produces an infeasible full-envelope controller, illustrating that compatibility labels alone do not establish information coverage.

VI. Supporting LQI Validation

Ten separate confirmation fleets (seeds 411–420) refit the maps and apply speed-scheduled LQI to nonlinear 12 s maneuvers with available physical-state feedback. At the eleven grid speeds, discrete Riccati synthesis uses $Q =$

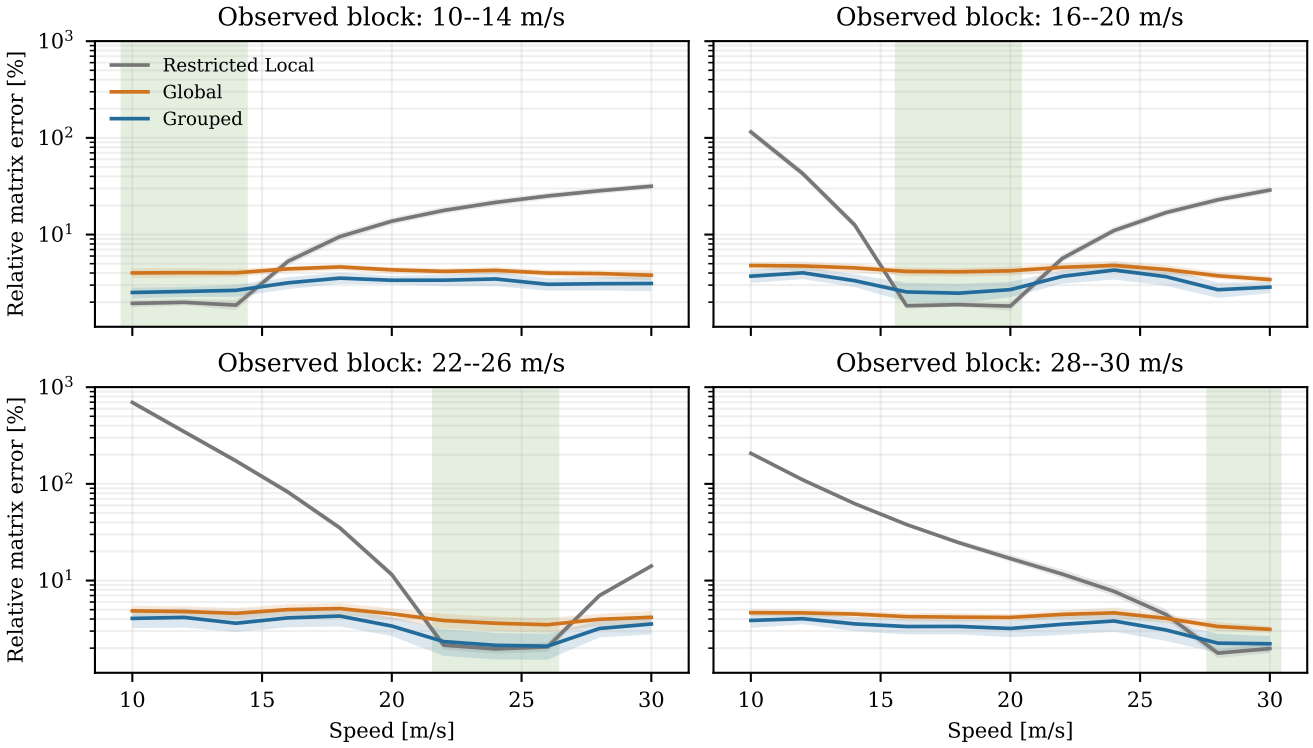


Fig. 2. Label-free matrix error by local coverage block. Shading marks observed speeds. Curves average within block and fleet, then across ten fleets, with pointwise 95% fleet-bootstrap bands. The logarithmic vertical axis retains both Local extrapolation failure and the smaller Global–grouped difference. The curves summarize the primary records; they are not a separate confirmation experiment.

$\mathrm{diag}(25, 50, 1000)$ for the two states and yaw-error integral, with steering weight $R = 0.15$. Gains and reference prefilters are linearly interpolated in speed. Weights were selected using only ExactLPV performance on development seeds 381–385, subject to feasibility and saturation checks. A weaker maneuver pilot on seeds 391–400 was discarded before the final confirmation. Neither grouped-versus-Global results nor confirmation fleets select weights.

The moderate maneuver smoothly varies speed $11 \to 29 \to 15$ m/s and uses an enveloped two-frequency lateral-acceleration demand with scale 4 m/s^2 and frequencies 0.35/0.72 Hz. The hard maneuver varies speed $29 \to 12 \to 28$ m/s and uses alternating Gaussian demands centered at 3.0/6.0/9.2 s, widths 0.40/0.48/0.55 s, and amplitudes $7/-7.7/5.6$ m/s^2. Yaw reference is requested acceleration divided by speed. Steering limits are 12° and 120°/s, with conditional integration, integral clipping at 0.6, and a 6° sideslip feasibility bound. These remain fixed across methods.

Mean paired tracking gains are 23.86%/8.96% against Local and 4.26%/1.21% against Global for moderate/hard demands. Global-relative intervals are $[1.95, 6.55]\%$ and $[0.47, 2.01]\%$. Absolute mean improvements are 0.001824/0.001874 degrees/s, so the control evidence supports usability with modest practical headroom. All federated trajectories satisfy the tested constraints. Maximum frozen-point spectral radius is 0.9759, which is not a scheduled or nonlinear stability certificate. Four fleets select one group, making federated and Global controllers identical there.

TABLE IV
SUPPORTING YAW TRACKING RMSE [DEGREES/S], AVERAGED ACROSS TEN CONFIRMATION FLEETS. OUTSIDE ITS OBSERVED BLOCK, LOCAL USES THE NEAREST-OBSERVED-SPEED CONTROLLER.

Method	Moderate	Hard
Restricted Local	0.0540	0.1698
Global	0.0429	0.1564
Federated groups	0.0411	0.1546
ExactLPV reference	0.0400	0.1507

VII. DISCUSSION AND CONCLUSION

The results distinguish complementary information, compatibility selection, and federation. Pooling speed blocks can supply directions absent from local regressions, but sharing across incompatible clients introduces bias. Grouped fitting improves full-envelope error over restricted Local, Regularized Local and Global while matching centralized fitting under the same records. The selected groups attain full scheduling rank in the evaluated fleets. Local regularization leaves the extrapolation error almost unchanged, supporting the role of complementary observations within the chosen LPV representation. The observed-speed advantage of Local also makes the tradeoff explicit: a shared map primarily addresses access across the operating interval.

The evidence is a controlled calibration proof of concept with simulator-derived centering and uncorrupted current regressors. It covers one cornering curve, not ordinary driving, noisy state estimates, real fleets, or frozen-library transfer. Group selection is heuristic, minimum size does not guarantee rank, and the exhaustive search incurs measurable communication. The observed full-rank coverage is an empirical result for these fleets, not a guarantee of the group-selection algorithm.

Within these assumptions, compatible federated regression enables clients with incomplete local coverage to obtain useful LPV maps over the speed interval. Nonlinear scheduled-LQI tests support model usability without requiring large tracking gains as the primary contribution.

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